

nanospice

A SPICE circuit simulator in 1000 lines of Rust

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github.com/milanofthe/nanospice

The rule

- one file, `src/main.rs`, at most 1000 nonblank, noncomment lines
- standard library only, no dependencies
- a test counts the lines and fails the build above the limit
- tests, comments, and example netlists are free

The budget forces every feature to displace another, which makes the cost of each decision explicit, and it keeps the simulator readable in one sitting.

current count: **992 of 1000**

What fits

Analyses

- `.op`, `.dc` sweep
- `.tran` with adaptive timestep, `uic`
- `.ac` dec/lin

Devices

- R, C, L, V, I, E, G
- diode, MOSFET, JFET, BJT
- junction and gate capacitances

Solver

- sparse LU with partial pivoting
- Newton with junction limiting
- gmin and source stepping
- trapezoid, LTE control, breakpoints

Input

- SIN, PULSE, PWL waveforms
- `.model`, `.print` cards

23 integration tests against analytic references, none of them counted

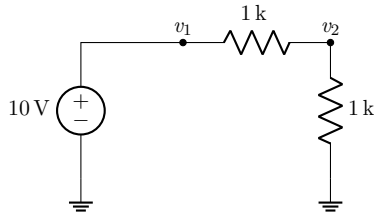
Where the lines go

code section	lines	cumulative
constants, error and output plumbing	23	23
numbers: Num trait, complex, unit suffixes	78	101
circuit data model	61	162
parser	275	437
sparse LU and stamp helpers	113	550
device models	64	614
solver: assembly, Newton, operating point	142	756
output selection	26	782
analyses and main	210	992

The parser is the largest section: the numerics of a circuit simulator are compact, the input format is not.

Modified nodal analysis

Unknowns: node voltages plus one branch current per voltage-defined element (V, L, E). KCL per node, one constraint row per branch.



$$\begin{pmatrix} g & -g & 1 \\ -g & 2g & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ i \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 10 \end{pmatrix}$$

$$v_1 = 10, \quad v_2 = 5, \quad i = -5 \text{ mA}$$

- the branch row has a structural zero on the diagonal: pivoting is mandatory
- $i < 0$: branch current flows into the positive terminal, a sourcing supply reads negative

The ground row

Design decision: a dummy ground row

Ground is unknown 0 and the solver loops start at 1. The alternative is an *if node is not ground* test inside every stamp, roughly two dozen branches across eleven devices. One guard inside `add` replaces all of them; $x_0 = 0$ is pinned after the solve.

```
fn quad(&mut self, p: usize, n: usize, cp: usize, cn: usize, g: T) {  
    self.add(p, cp, g);  
    self.add(p, cn, T::zero() - g);  
    self.add(n, cp, T::zero() - g);  
    self.add(n, cn, g);  
}
```

The conductance `quad` covers R and the VCCS; `branch` covers V, L, and E. Every stamp in the file reduces to these two shapes.

Sparse LU: the data structure

sparse LU and stamp helpers

113 lines, budget 437 to 550



- each matrix row: a vector of (column, value) pairs, sorted by column
- eliminated columns are removed from active rows

The first-entry invariant

Before elimination step k , every row stored at index $i \geq k$ contains only columns $\geq k$. Consequence: if such a row has an entry in column k at all, it is the *first* element of its vector. Pivot search is an $O(1)$ probe per candidate row, with no column-linked index structure.

Sparse LU: elimination is a merge

```
// row a minus f times pivot row p, both sorted
fn merge<T: Num>(a: &[(usize, T)], p: &[(usize, T)], f: T) -> Vec<(usize, T)> {
    let (mut i, mut j) = (0, 0);
    let mut o = Vec::with_capacity(a.len() + p.len());
    while i < a.len() && j < p.len() { /* three-way column compare */ }
    // fill-in falls out of the merge as the union of the column sets
    o
}
```

- pivoting: largest magnitude among the candidate first entries, rows swapped whole
- no candidate for column k : matrix is singular, the failing column is named singular matrix at node b , no conduction path?
- no Markowitz ordering: netlist order, linear cost on ladders and trees
- 2000-node RC ladder, OP plus AC sweep: tens of milliseconds

One primitive for every nonlinear device

A static device model is a set of current branches, each controlled by node-pair voltages. This is the contribution operator of Verilog-A, $I(p, n) <+ f(v)$:

```
fn contrib(&mut self, p: usize, n: usize, i: T,
          ctl: &[(usize, usize, T)], x: &[T]) {
    let mut ieq = i;
    for &(a, b, g) in ctl {
        self.quad(p, n, a, b, g);
        ieq = ieq - g * (x[a] - x[b]);
    }
    self.rhs(p, T::zero() - ieq);
    self.rhs(n, ieq);
}
```

The dual `vcontrib` writes the branch row $v_p - v_n - \sum_k g_k(x_{a_k} - x_{b_k}) = v$ for the voltage-defined elements.

The whole device set on two primitives

device	primitive	controls
R, C, I	contrib	self conductance, none for I
G (VCCS)	contrib	one control pair
diode	contrib	one junction
MOSFET, JFET	contrib	g_m and g_{ds}
BJT	$3 \times \text{contrib}$	two junctions, transport source
V	vcontrib	none
L	vcontrib	its own branch current
E (VCVS)	vcontrib	one control pair

The refactor to this form removed more lines than the primitives cost; the freed budget paid for the JFET and the PWL source. Hand-derived Jacobian stamps become verification artifacts.

Diode and junction limiting

Newton diverges on the bare exponential: an early iterate proposing 5 V puts e^{193} into the Jacobian.

$$v_{crit} = V_T \ln \frac{V_T}{\sqrt{2} I_S} \quad (\text{the point of maximum curvature of } i(v))$$

Beyond v_{crit} , `pnjlim` replaces the proposed step by a logarithmic one:

$$v \leftarrow v_{old} + V_T \ln \left(1 + \frac{v_{new} - v_{old}}{V_T} \right)$$

- the junction advances a controlled amount per iteration, regardless of overshoot
- per-device memory of the last limited voltage; the BJT carries two

Limiting must deny convergence

Field note

With limiting alone, a common-emitter amplifier converged with the transistor off: the divider put the base at 2.1 V, `pnjlim` clamped the junction seen by the stamps to 0.11 V, no current flowed, the node voltages stopped changing, and the delta test declared success. The output was superficially plausible: the divider voltage was correct, but unloaded.

- repair: an iteration in which a limiter changed a junction voltage cannot converge
- the assembly reports a `limited` flag, the Newton loop requires it false
- with the flag: correct bias (2.05 V, 1.3 mA) in a dozen iterations
- a convergence test on the solution vector alone cannot detect active limiting

MOSFET: the effective frame

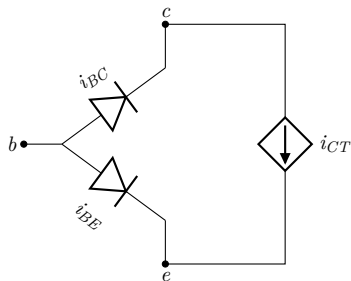
Four cases (nmos/pmos, source-drain swap), one evaluation. With polarity $s = \pm 1$ and effective terminals chosen so $v_{DS}^e \geq 0$:

$$\frac{\partial I}{\partial v_g} = s \cdot g_m \cdot s = g_m$$

Every derivative picks up the polarity twice, and $s^2 = 1$: the conductance stamps are identical for all four cases, only the right-hand side carries s .

- square law with channel-length modulation, g_m and g_{ds} by differentiation
- the JFET is the same square law with $K_P = 2\beta$ and a depletion threshold: it shares the MOSFET variant and parser arm

BJT: three contributions



$$i_{CT} = I_S(e_f - e_r), \quad i_{BE} = \frac{I_S}{\beta_F}(e_f - 1), \quad i_{BC} = \frac{I_S}{\beta_R}(e_r - 1)$$

```
s.contrib(*b, *e, ..., &[(*b, *e, gpi)], x);  
s.contrib(*b, *c, ..., &[(*b, *c, gmu)], x);  
s.contrib(*c, *e, ...,  
          &[(*b, *e, gmf), (*b, *c, -gmr)], x);
```

- KCL composes the classic 3×3 stamp; rows and columns sum to zero by construction
- npn through the same sign argument; an npn/pnp mirror test requires exact symmetry

Junction capacitances by desugaring

A capacitance across a node pair is exactly the `C` device, so the parser desugars `cjo`, `cje/cjc`, `cgs/cgd` into internal capacitors. No assembly, state, timestep, or AC code knows they exist.

$$C(v) = C_{j0} \left(1 - \frac{v}{V_J}\right)^{-m}, \quad Q(v) = \frac{C_{j0} V_J}{1 - m} \left(1 - \left(1 - \frac{v}{V_J}\right)^{1-m}\right)$$

- charge-based capacitor, linearized above $F_C V_J$; $m = 0$ is the plain linear `C`
- junctions graded ($m = 0.5$), gate oxide linear, `m=` available on the `C` card
- test: at $v = -3\text{ V}$ the capacitance is exactly $C_{j0}/2$, the AC corner doubles

Operating point: homotopy paths as data

solver: assembly, Newton, operating point

142 lines, budget 614 to 756

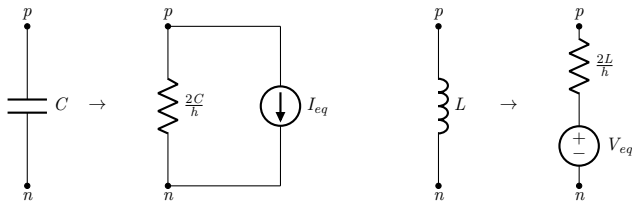


Both classic fallbacks are paths in the $(g_{min}, \text{source scale})$ plane, walked with warm starts:

```
let gmin_path: Vec<(f64, f64)> =  
    (2..=12).map(|k| (10f64.powi(-k), 1.0)).collect();  
let src_path: Vec<(f64, f64)> =  
    (1..=10).map(|k| (GMIN, k as f64 / 10.0)).collect();  
for path in [gmin_path, src_path] {  
    if path.iter().all(|&(g, fac)|  
        newton(ckt, x, lim, &Mode::Dc { fac }, g, ITL_OP).is_some()) {  
        return;  
    }  
}
```


Transient: companion models

Trapezoidal rule on $i = C dv/dt$: the capacitor becomes a resistor with a history source for one step. The inductor is the dual through the flux $\varphi = Li$.



- state per reactive device: charge or flux plus the dual rate, one update rule for both
- trapezoid: second order, A-stable, no numerical damping (an LC tank holds its amplitude for twenty periods, a test enforces it)

Timestep control by truncation error

$$\varepsilon_{tr} = -\frac{h^3}{12} \ddot{x}, \quad \varepsilon_{tr} \approx \frac{h^3/12}{h(h+d_1)(h+d_1+d_2)/6 + h^3/12} (x_c - x_p)$$

- quadratic predictor through the last three accepted points, Lagrange coefficients for variable steps
- corrector minus predictor measures $\ddot{x}$ (Milne estimate); uniform steps give the classic factor 1/13
- cube-root step rule $h' = 0.9 h r^{-1/3}$, clamped to $[0.3, 2] \times h$
- the predictor doubles as the Newton starting point
- nonconvergence: cut the step by 8, the SPICE2 default

Breakpoints and damped restart

A pulse edge inside a step is invisible: the companion history belongs to the pre-edge circuit. For $h/\tau = 0.1$:

$$v_1 = \frac{h/2\tau}{1 + h/2\tau} \approx 0.048 \quad \text{instead of} \quad 1 - e^{-h/\tau} \approx 0.095$$

a factor-two error no smaller h fixes.

- every point of every piecewise linear waveform is a breakpoint; steps land on them exactly
- after a breakpoint: small step, cleared predictor history
- the two predictor-less bootstrap steps run backward Euler: first order but strongly damped, absorbing the trapezoid ringing at the discontinuity

Small-signal AC

$$(G + j\omega C) \hat{x} = \hat{b}$$

At a converged operating point the limiters pass voltages through unchanged, so G is *exactly the DC Jacobian the last Newton iteration assembled*. The AC path reuses `assemble`, maps to complex, adds $j\omega C$ and $-j\omega L$.

```
trait Num: Copy + Add<Output = Self> + Sub<Output = Self>
  + Mul<Output = Self> + Div<Output = Self> {
  fn zero() -> Self;
  fn one() -> Self;
  fn mag(self) -> f64;    // pivot magnitude: abs for f64, modulus for Cx
}
```

One generic sparse LU serves Newton and AC; no device model contains any AC-specific code.

It oscillates

```
cmos ring oscillator
.model n nmos kp=1e-3 vt0=1 cgs=5p cgd=2p
.model p pmos kp=1e-3 vt0=1 cgs=5p cgd=2p
vdd vdd 0 dc 5
m1 b a 0 0 n
m2 b a vdd vdd p
m3 c b 0 0 n
m4 c b vdd vdd p
m5 a c 0 0 n
m6 a c vdd vdd p
i1 0 a pulse(0 1m 0 1n 1n 20n)
.tran 1n 500n
```

- three inverters, gate capacitances set the stage delay
- current kick leaves the metastable operating point
- period 78 ns, stable within 1.5 %
- a test measures the period and bounds the jitter

Without the capacitances the transistors switch instantaneously and the ring has no defined frequency.

Validation

tests/cli.rs

330 lines, 0 of 992: tests are free



References independent of the implementation:

- analytic: RC/RL step vs. $1 - e^{-t/\tau}$, AC corner at $1/\sqrt{2}$ and -45° , LC amplitude $i_0\sqrt{L/C}$, energy conservation
- bias points: MOSFET, JFET, BJT ($I_C = \beta_F I_B$), exact npn/pnp symmetry
- structural: a randomized 30-stage ladder verified against a Thevenin reduction computed independently in the test
- negative: seven malformed decks must produce named errors, never a panic; the floating node error must name the node
- meta: the line counter itself

The cut list

feature	why it lost	est. lines
<code>.subckt</code> hierarchy	lost to the transistor models	~100
Markowitz ordering	netlist order suffices at this scale	~80
Gummel-Poon BJT	Ebers-Moll covers the teaching core	~80
noise analysis	needs the adjoint system	~80
<code>.param</code> expressions	a calculator is its own project	~60
temperature sweeps	V_T is a constant, by choice	~25

The most consequential remaining cut is `.subckt`: real decks are hierarchical. It would be the first addition if the budget grew.

992 of 1000 lines

MNA, sparse LU with partial pivoting, two contribution primitives carrying all device stamps, Newton with pnjlim and data-driven homotopy fallbacks, trapezoidal integration with LTE control, breakpoints and damped restarts, charge and flux based reactive state with graded junction capacitances, small-signal AC, eleven device letters, four analyses, 23 tests.

`github.com/milanofthe/nanospice`

MIT licensed; the design report with all derivations is in the repository.